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Paper title: Electromyography-Based Hand Gesture Recognition System for Upper Limb Amputees

Paper authors: Sidharth Pancholi and Amit M. Joshi

Research field of the orientative paper:

Using surface electromyography (EMG) signals of amputees to recognize their arm gestures.

Summary of the paper:

This paper presents a surface electromyography (EMG) signal-based system for real-time recognition of arm gestures in amputees. The system is capable of recognizing user intention in the real-time environment, with above 91% accuracy.

Contribution (State of the Art Comparison)

- Designed a flexible and adjustable armband for forearm EMG signal acquisition.
- Proposed a system capable of online training for users and predicting the subject's transient intentions.
- Developed a real-time embedded platform capable of detecting user intentions within 75 milliseconds, with an accuracy above 91%.

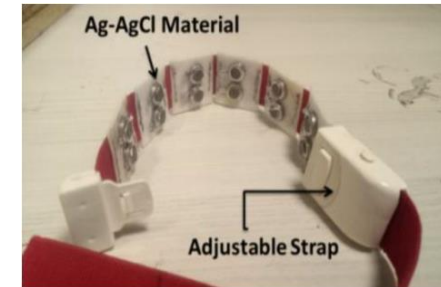
Results presented in the paper

- Designed an armband that can acquire 8-channel EMG signals simultaneously.

- Accuracy:

In Case 1 (feature extraction in DSP processor and testing off-line), the classification accuracy reached up to 97.6% for healthy subjects, while the accuracy was 92% for the amputee.

In Case 2 (training and testing within DSP processor), the overall mean classification accuracy was 94.14%; for healthy subjects, the highest classification accuracy reached 97.75%, while for the amputee, it was 91.65%.



Critical reasoning about the results explained in the paper

- The flexible and adjustable armband could effectively reduce interference to the subjects during EMG signal acquisition, thereby improving the stability and naturalness of signal collection.
- The system also has great potential for assisting non-amputees with limited mobility, such as injured individuals and the elderly.
- Applying machine learning or deep learning models to the acquired EMG signals could potentially further enhance the accuracy of classification and prediction.